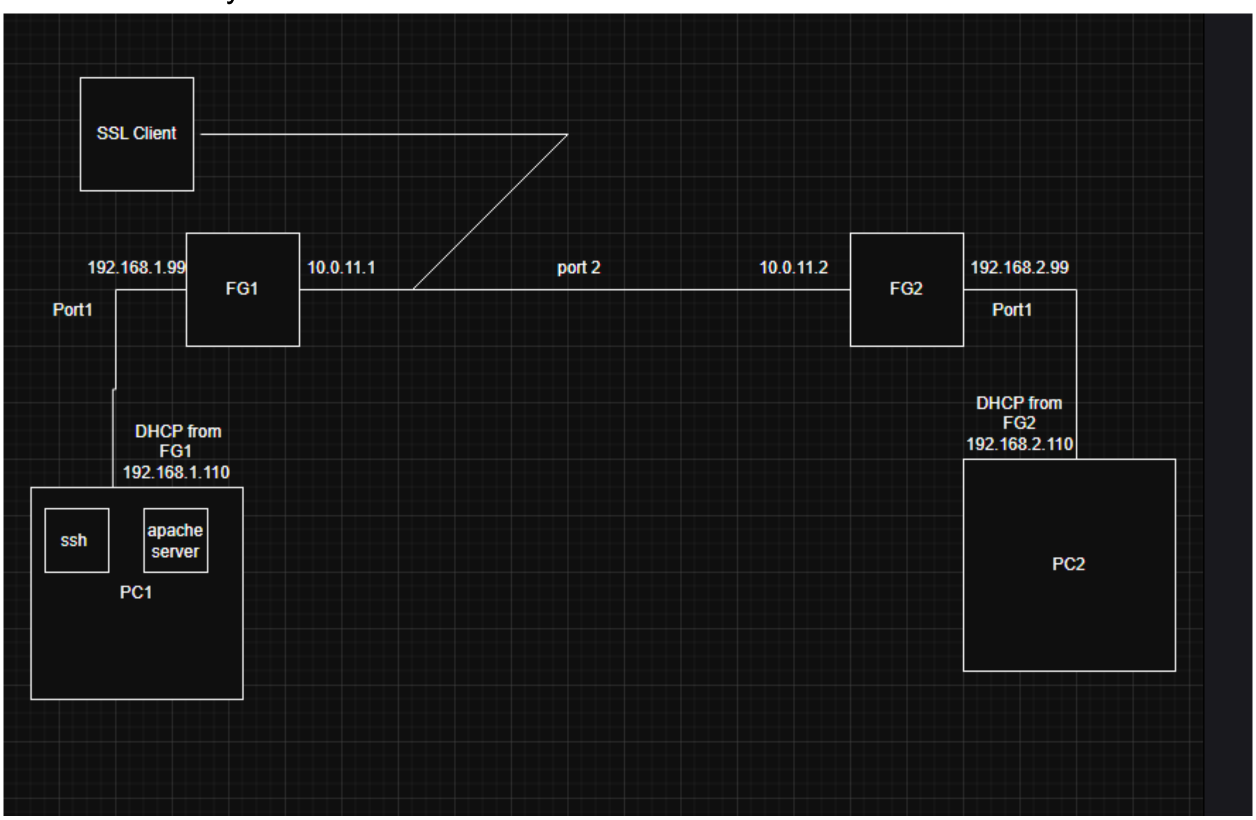
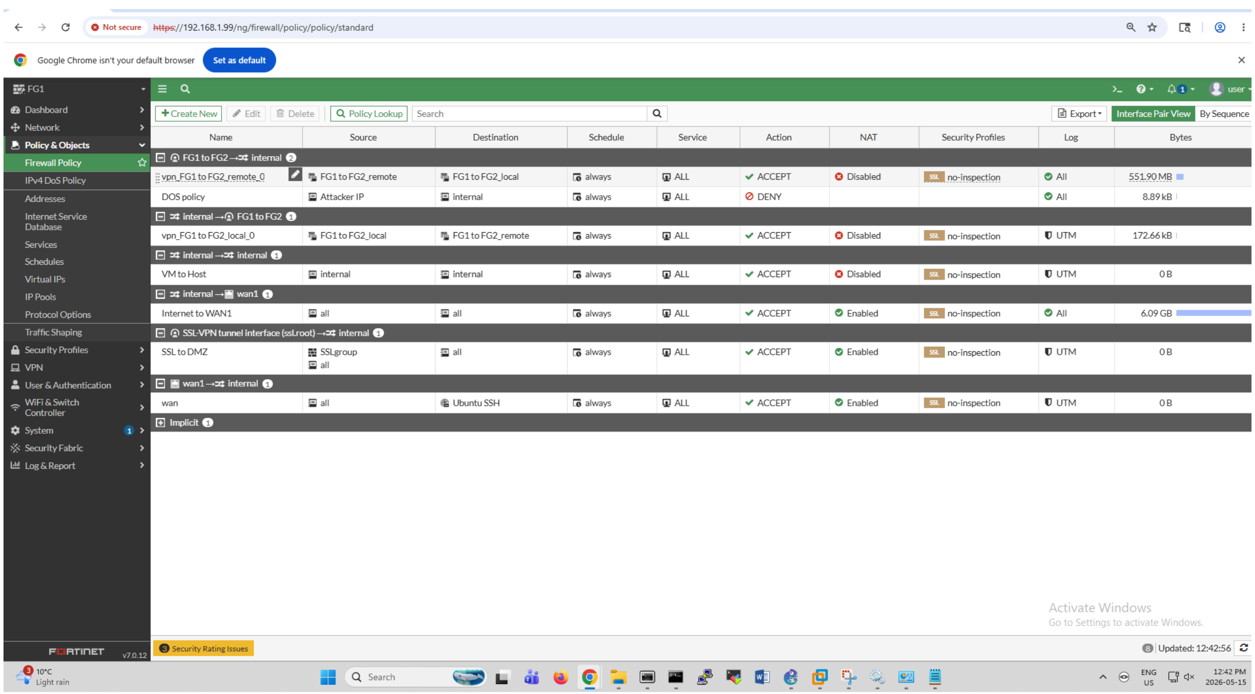
The network design features two core geographic locations interconnected across a transit interface, along with provisioning for a secure remote workforce. The primary internal headquarters LAN is built out behind a Fortinet FortiGate appliance, designated as FG1, which serves as the default gateway for PC1. Within this primary subnet, PC1 is connected to FG1 with an IP address of 192.168.1.99 for out-of-band management and direct administrative access to the firewall's graphical user interface. To facilitate dynamic host configurations for standard network endpoints, a Local Area Network (LAN) Dynamic Host Configuration Protocol (DHCP) server was configured on FG1 to distribute addresses within the 192.168.1.1 to 192.168.1.255 range. To maintain administrative access and prevent address collisions with static assets, an exclusion pool was strictly defined from 192.168.1.90 through 192.168.1.99.

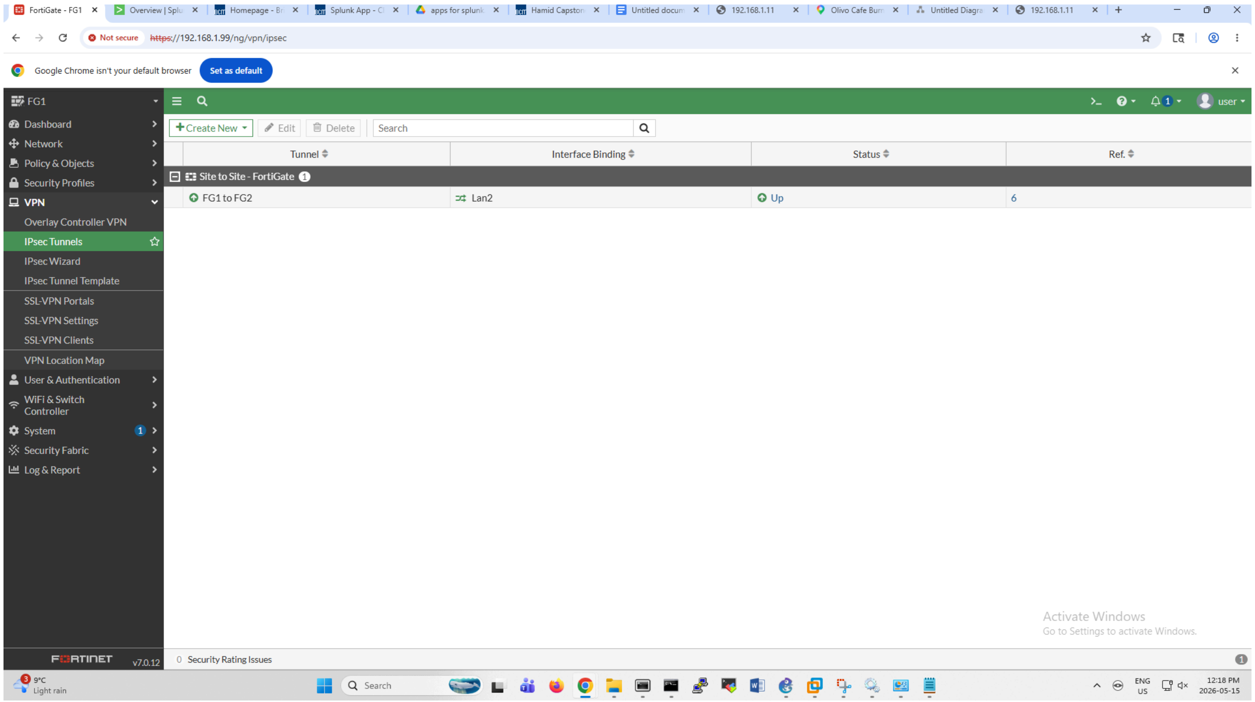
the secondary branch location operates under a mirror topology managed by a second Fortinet appliance, designated as FG2. For this site, PC2 was configured with an IP address of 192.168.2.99. The Port 1 address on FG2 was modified to act as the internal gateway interface, changing its address space to correctly bind to the secondary LAN network. The local DHCP scope on the FG2 port was subsequently adjusted to hand out the appropriate matching IP network range, enabling newly connected branch hosts to successfully acquire dynamic IP leases, resolve their default gateways, and achieve local connectivity.

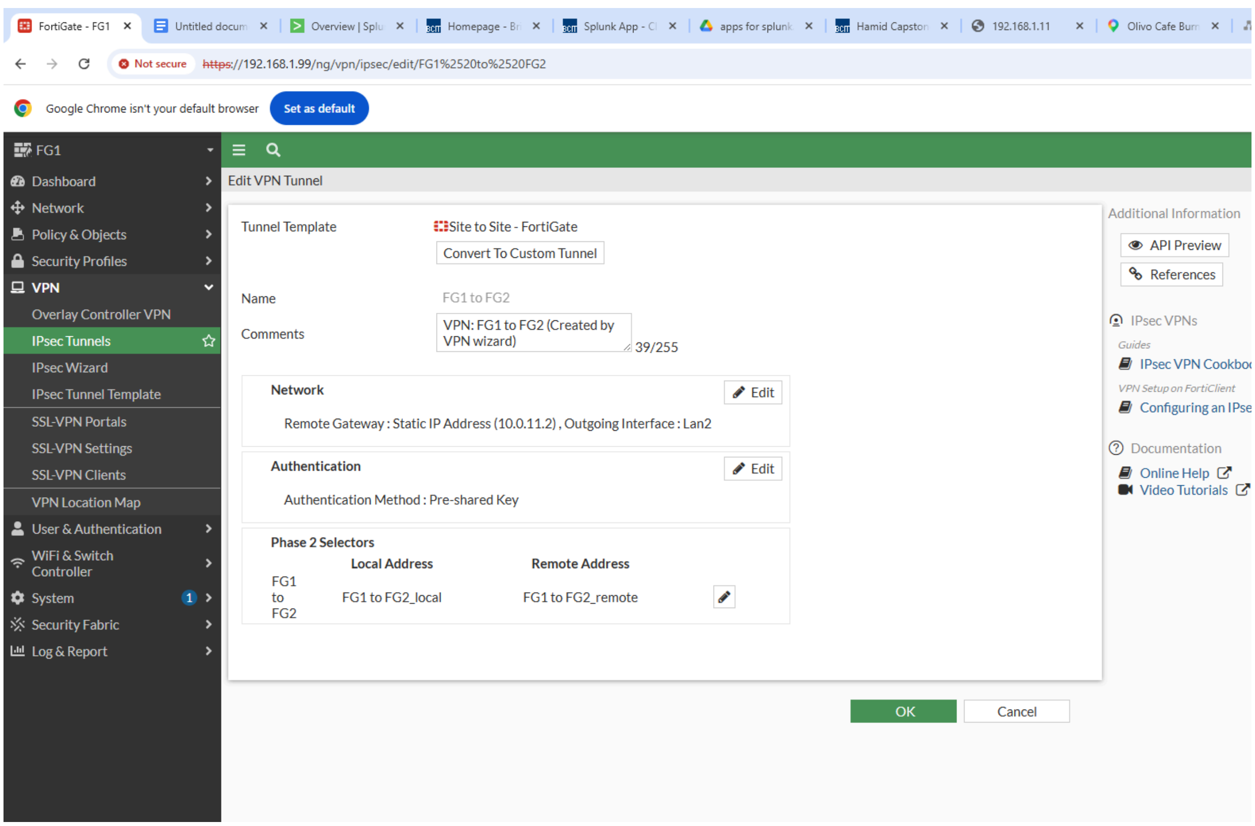


Implement Site-to-Site VPN

A Site-to-Site IPsec Virtual Private Network tunnel was established between the appliances. The interconnect layout leverages Port 2 on both firewalls as the dedicated outbound public-facing WAN interface for the remote IP outgoing interface connection. During the network design phase, FG1 was assigned a local IP of 10.10.11.1 and FG2 was assigned a local IP of 10.10.11.2. For security policy definition, the remote subnet configuration for FG1 was mapped to the branch network 192.168.2.0, while the remote subnet configuration for FG2 was mapped back to the primary HQ network 192.168.1.0 using the Internet Key Exchange version 1 (IKEv1) To bring the tunnel online, active ICMP echo requests IPsec Monitor VPN tunnel dashboard, the connection was brought up manually by activating it, forcing an active status change and successfully permitting encrypted bidirectional traffic flow between the two networks

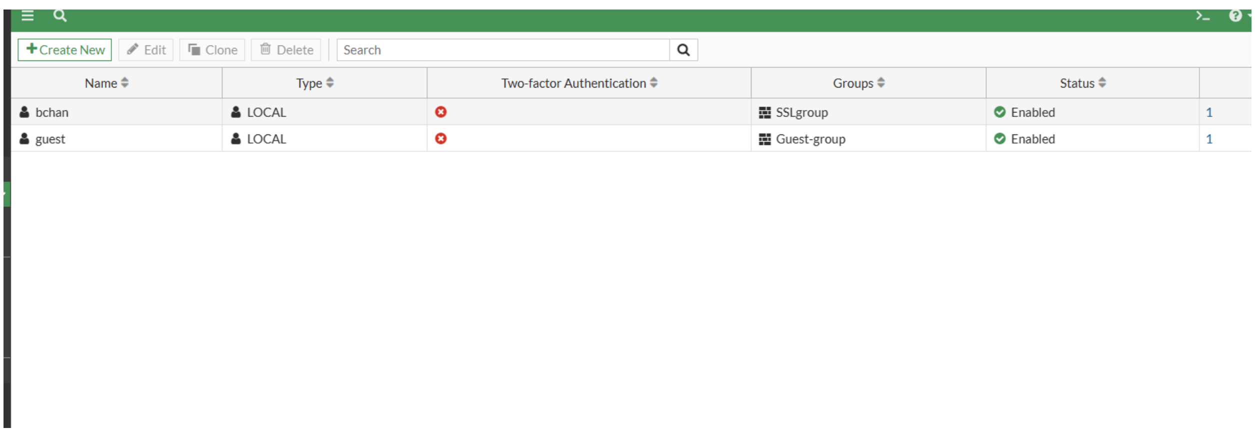


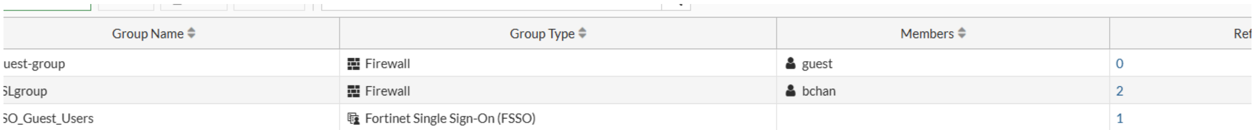




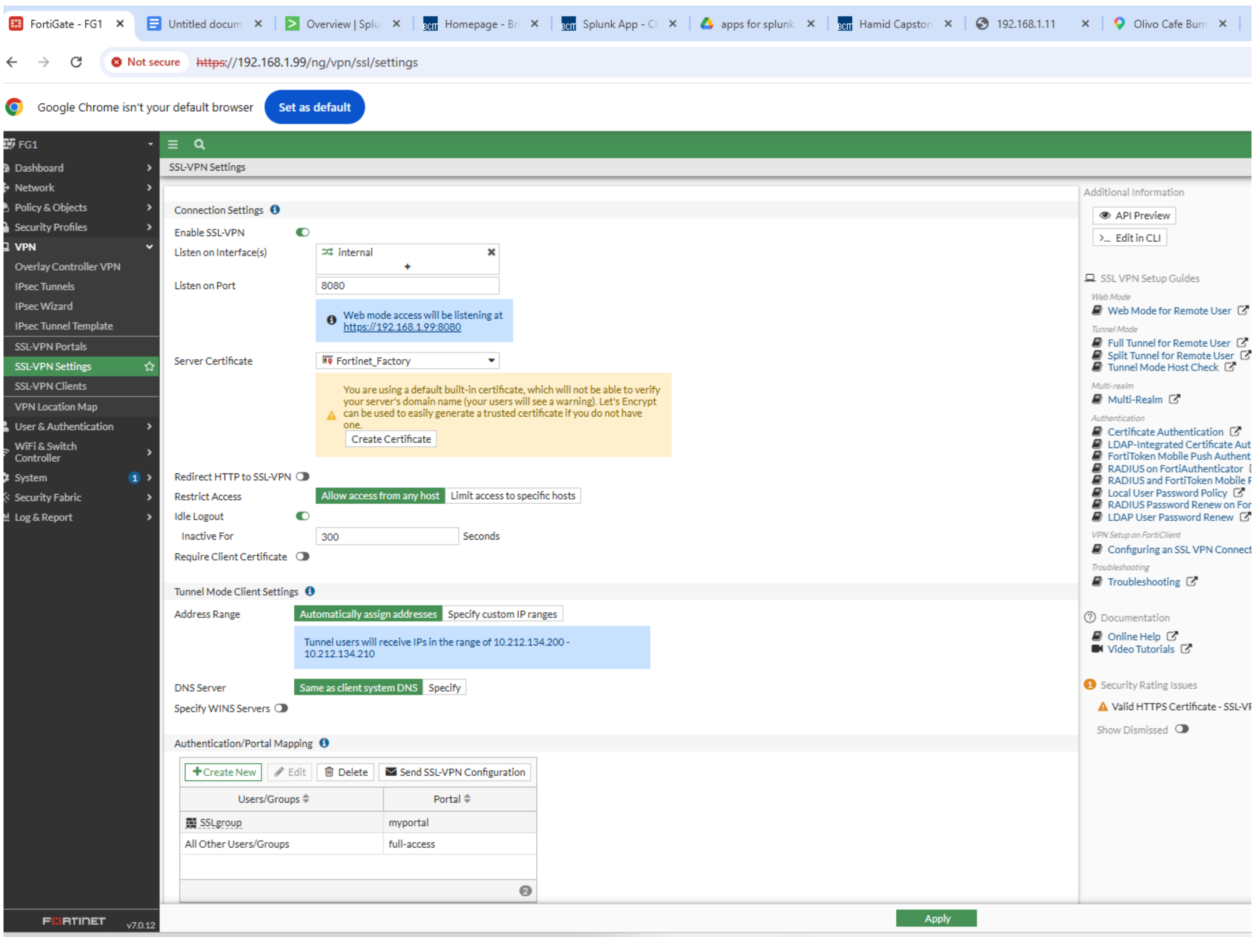
Configure Client-Based VPN for remote access

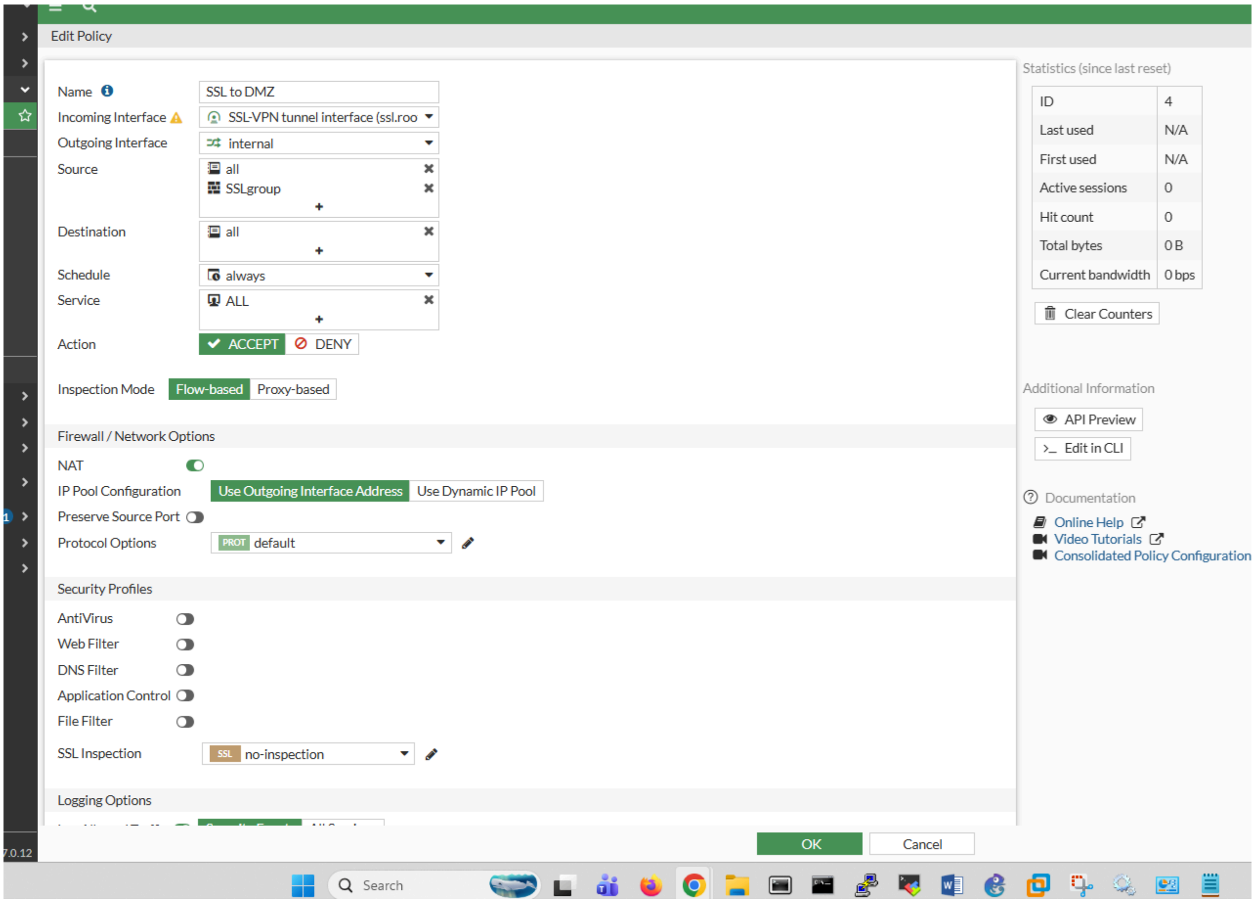
remote employee connectivity, a client-based Secure Sockets Layer (SSL) VPN was implemented on the primary firewall ,a Windows-based virtual machine was deployed inside PC1 with client user being assigned a BCIT address An external user account bchan was first created to handle remote user login authentication This user account was added to the correct user group policy SSLgroup An SSL-VPN portal was then created on listening port 8080, and configured to assign users a specific IP pool when connecting to the tunnel. The user group was added to the authentication mapping parameters during the SSL-VPN setup When configuring the stateful SSL policy on FG1, the source was assigned to the SSL group, allowing all incoming connections to route to internal targets Remote users are required to download the Fortinet VPN client software to connect to the internal network remote gateway setting to 192.168.1.99 over custom port 8080, and using the created user account and correct password, the user successfully connected to the internal firewall and verified network visibility by pinging internal users

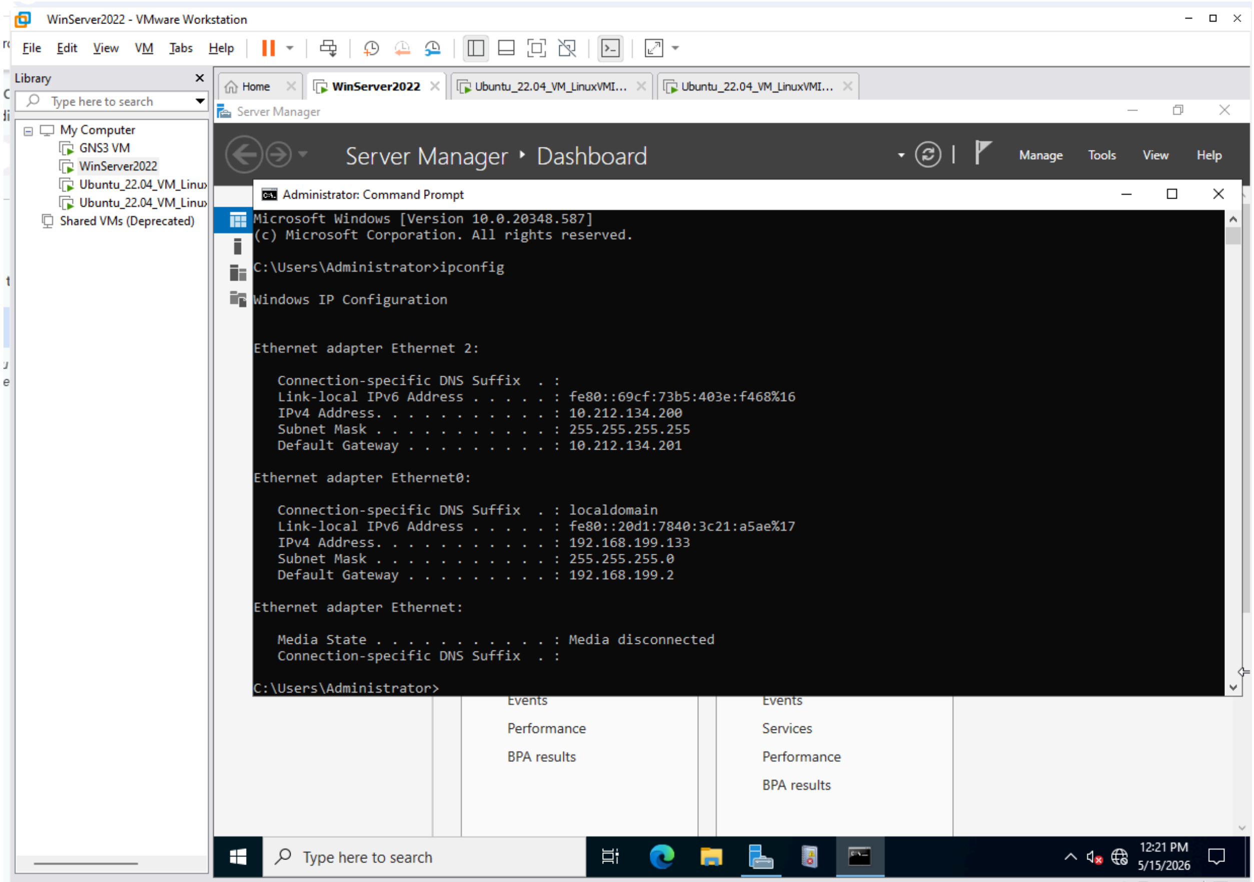


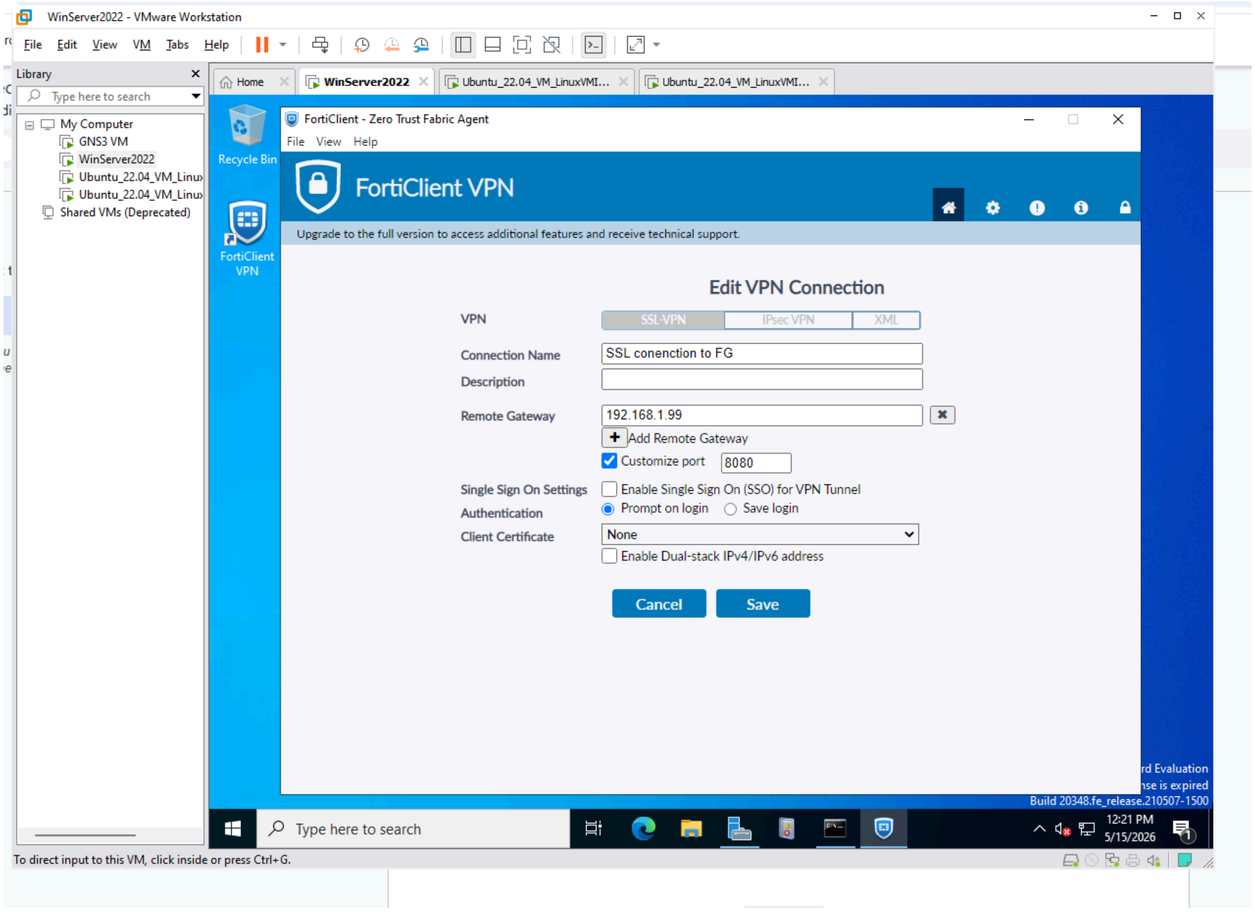


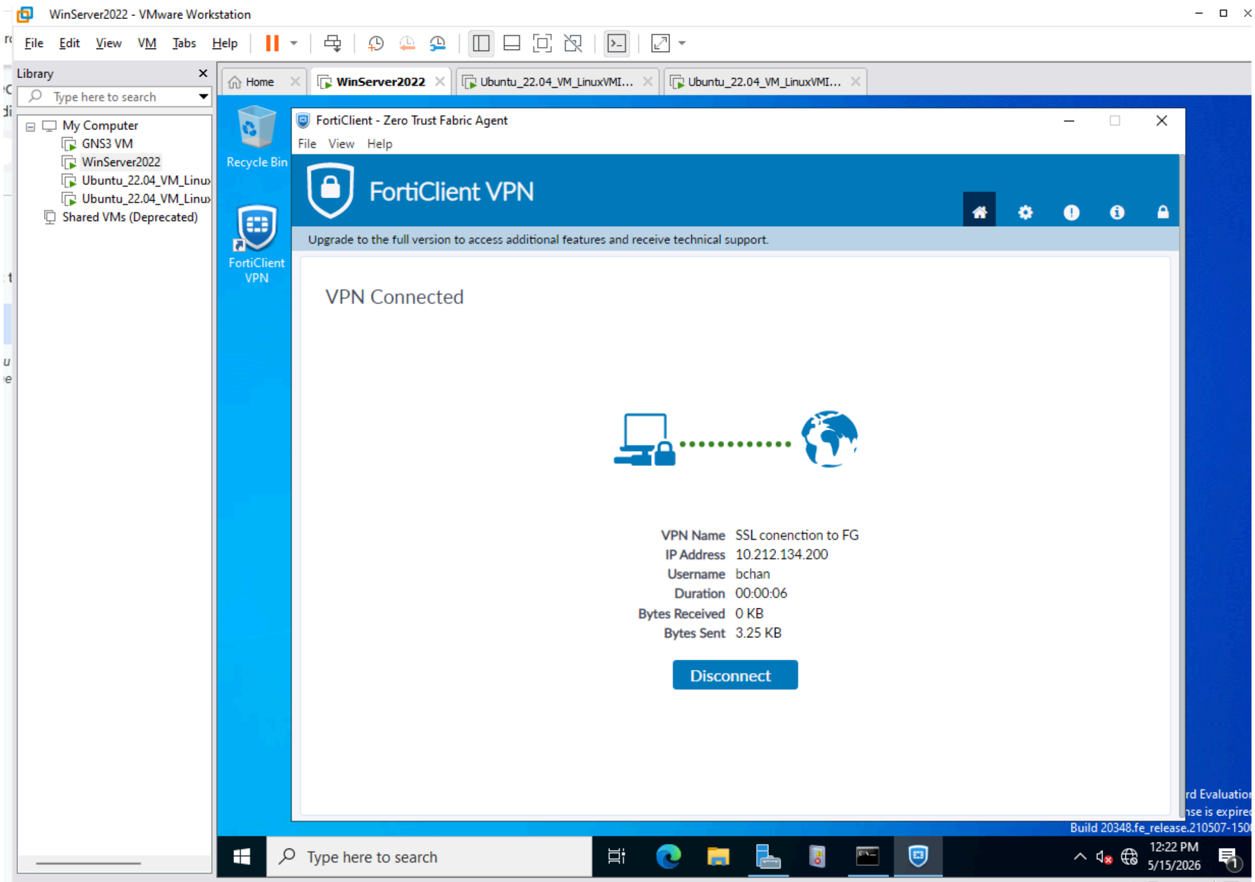
Group Name	Group Type	Members	Ref
uest-group	Firewall	guest	0
SLgroup	Firewall	bchan	2
SO_Guest_Users	Fortinet Single Sign-On (FSSO)		1

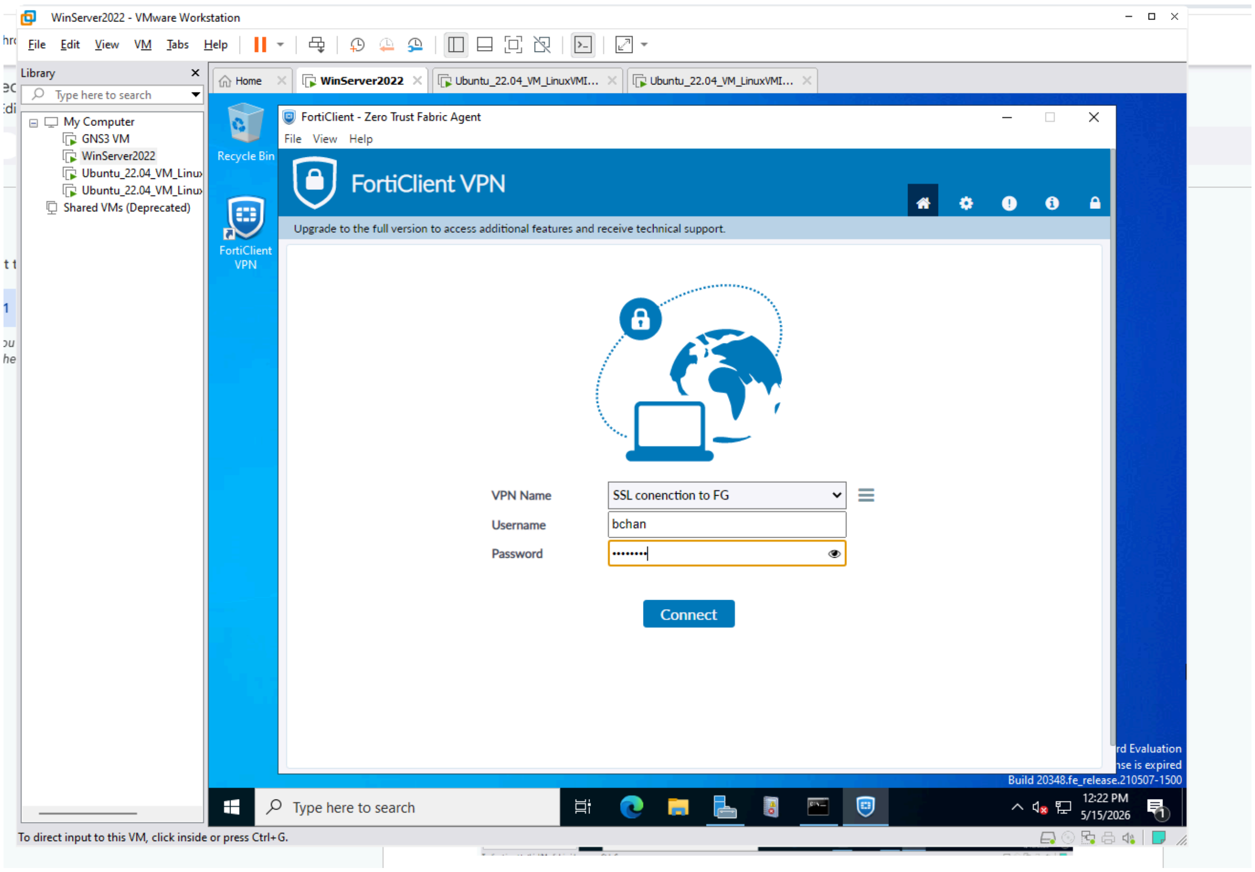



Edit Policy
Name SSL to DMZ
Incoming Interface SSL-VPN tunnel interface (ssl.roo
Outgoing Interface internal
Source all
SSLgroup
Destination all
Schedule always
Service ALL
Action ACCEPT DENY
Inspection Mode Flow-based Proxy-based
Firewall / Network Options
NAT
IP Pool Configuration Use Outgoing Interface Address Use Dynamic IP Pool
Preserve Source Port
Protocol Options PROT default
Security Profiles
AntiVirus
Web Filter
DNS Filter
Application Control
File Filter
SSL Inspection SSL no-inspection
Logging Options
OK
Cancel
Statistics (since last reset)
ID 4
Last used N/A
First used N/A
Active sessions 0
Hit count 0
Total bytes 0 B
Current bandwidth 0 bps
Clear Counters
Additional Information
API Preview
Edit in CLI
Documentation
Online Help
Video Tutorials
Consolidated Policy Configuration
7.0.12
Search


WinServer2022 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
GNS3 VM
WinServer2022
Ubuntu_22.04_VM_Linux
Ubuntu_22.04_VM_Linux
Shared VMs (Deprecated)
Home
WinServer2022
Ubuntu_22.04_VM_LinuxVMI...
Ubuntu_22.04_VM_LinuxVMI...
Server Manager
Server Manager ‣ Dashboard
Manage Tools View Help
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet 2:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::69cf:73b5:403e:f468%16
IPv4 Address. . . . . : 10.212.134.200
Subnet Mask . . . . : 255.255.255.255
Default Gateway . . . . : 10.212.134.201
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . : localdomain
Link-local IPv6 Address . . . . . : fe80::20d1:7840:3c21:a5ae%17
IPv4 Address. . . . . : 192.168.199.133
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . : 192.168.199.2
Ethernet adapter Ethernet:
Media State . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\Administrator>
Events
Performance
BPA results
Events
Services
Performance
BPA results
Type here to search
12:21 PM
5/15/2026


WinServer2022 - VMware Workstation
FortiClient - Zero Trust Fabric Agent
FortiClient VPN
Upgrade to the full version to access additional features and receive technical support.
Edit VPN Connection
VPN
SSL-VPN
IPsec VPN
XML
Connection Name
SSL conenction to FG
Description
Remote Gateway
192.168.1.99
Add Remote Gateway
Customize port
8080
Single Sign On Settings
Enable Single Sign On (SSO) for VPN Tunnel
Authentication
Prompt on login
Save login
Client Certificate
None
Enable Dual-stack IPv4/IPv6 address
Cancel
Save
To direct input to this VM, click inside or press Ctrl+G.

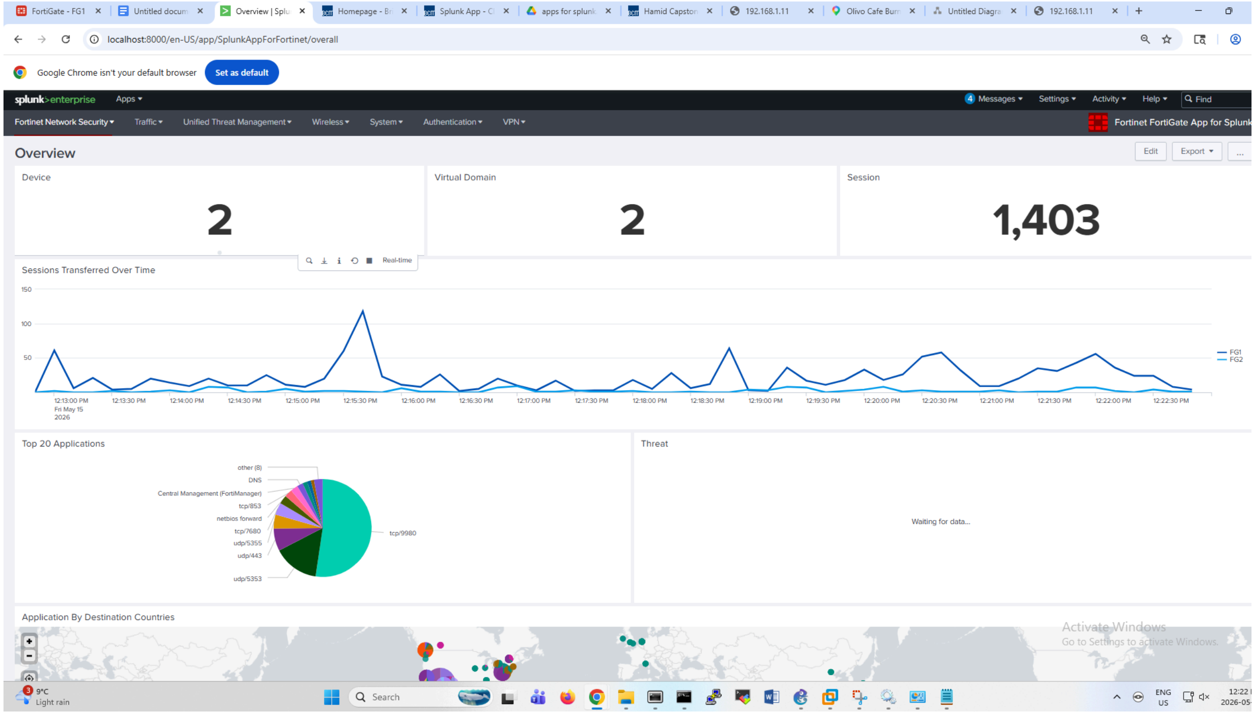

WinServer2022 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
GNS3 VM
WinServer2022
Ubuntu_22.04_VM_Linux
Ubuntu_22.04_VM_Linux
Shared VMs (Deprecated)
Home
WinServer2022
Ubuntu_22.04_VM_LinuxVMI...
Ubuntu_22.04_VM_LinuxVMI...
Recycle Bin
FortiClient VPN
FortiClient - Zero Trust Fabric Agent
File View Help
FortiClient VPN
Upgrade to the full version to access additional features and receive technical support.
VPN Connected
VPN Name SSL conenction to FG
IP Address 10.212.134.200
Username bchan
Duration 00:00:06
Bytes Received 0 KB
Bytes Sent 3.25 KB
Disconnect
Type here to search
12:22 PM
5/15/2026

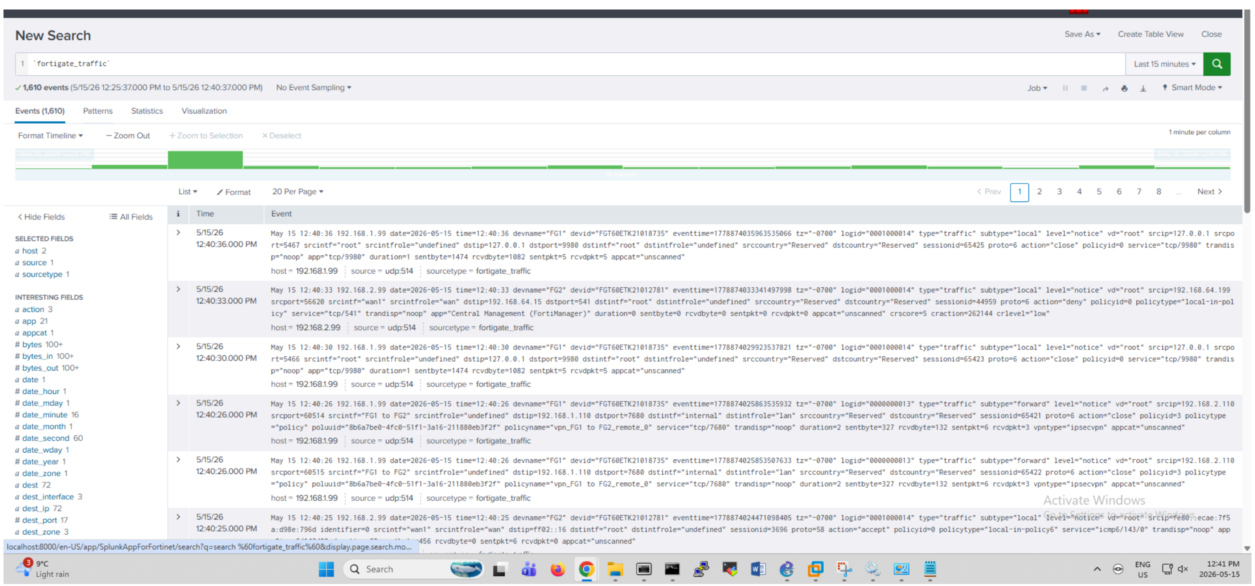


Send all logs to Splunk
two attack SSH bruteforce attack and an Nmap port scan were simulated inside the environment using an Kali Linux Kali Linux virtual machine was built on PC1, and a secondary Kali Linux virtual machine was created inside PC2 to simulate an internal attack vector via SSH
After ensuring baseline network routing by verifying that the PC2 Kali Linux VM was able to successfully ping the PC1 Kali Linux VM, an brute force attack was launched. From the PC2 virtual machine, the login cracker Hydra was executed against the internal host target 192.168.1.10 hydra -l ubuntu -P pass.txt 192.168.1.10 ssh -vV
shown in the documentation screenshots, Splunk successfully tracked the traffic lifecycle of the brute-force The logs provide evidence that the host at 192.168.2.10 successfully completed an unauthorized SSH authentication sequence into the target server at 192.168.1.10
The Kali Linux VM ciphered through the entries in the supplied password dictionary file pass.txt against the target host on PC1 It eventually succeeded in

finding the correct login credentials, allowing the attacker to authenticate and log in via SSH using the command ssh ubuntu@192.168.1.10

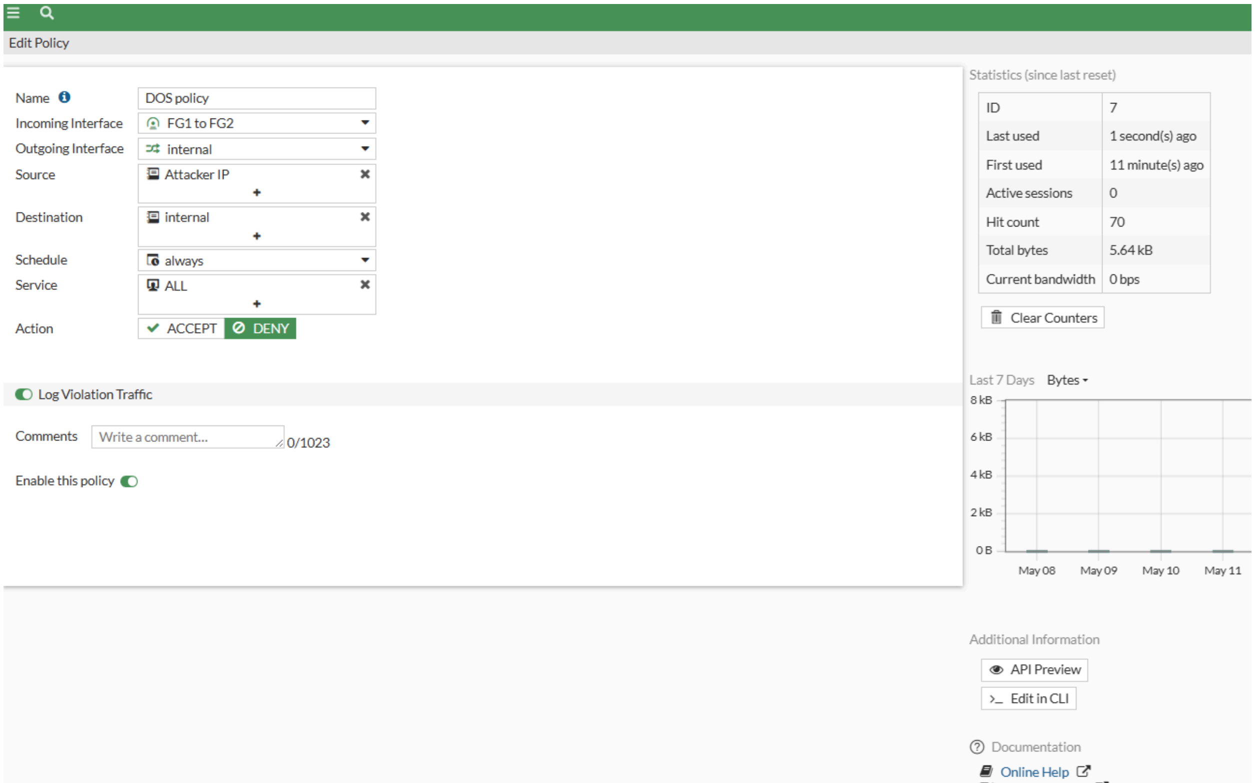
The Nmap network utility was run against the target system using service-version detection flags nmap -sV 192.168.1.10 The scan queried standard destination transport ports to check all available and open ports for access. The output demonstrated that an SSH service was openly available on the host, along with an HTTP service running on IP 192.168.1.11, which hosts the company's internal Apache web server.event logs further showed that the attacker at 192.168.2.10 attempted to scan the network for active ports like SSH and HTTP However, a pre-configured DoS Protection Policy set up on the firewall actively monitored these inbound scanning sweeps. The policy flagged the rapid connection spikes as malicious scanning anomalies and successfully blocked the incoming scans from executing fully.

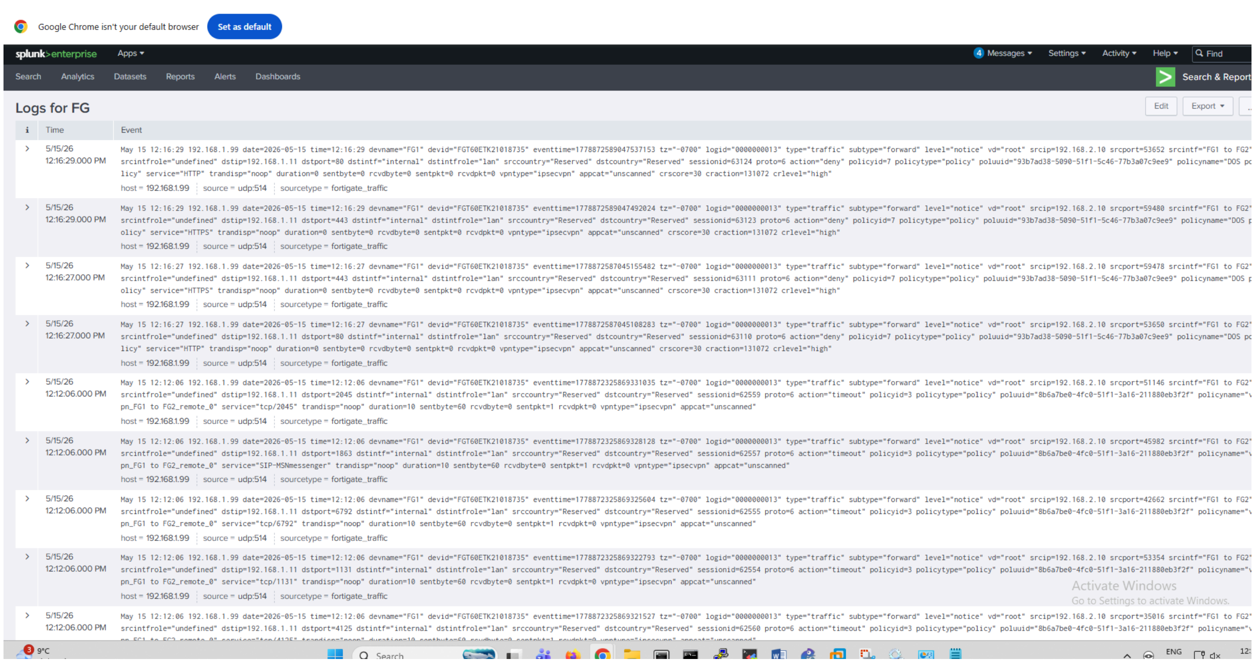




Logging & Splunk

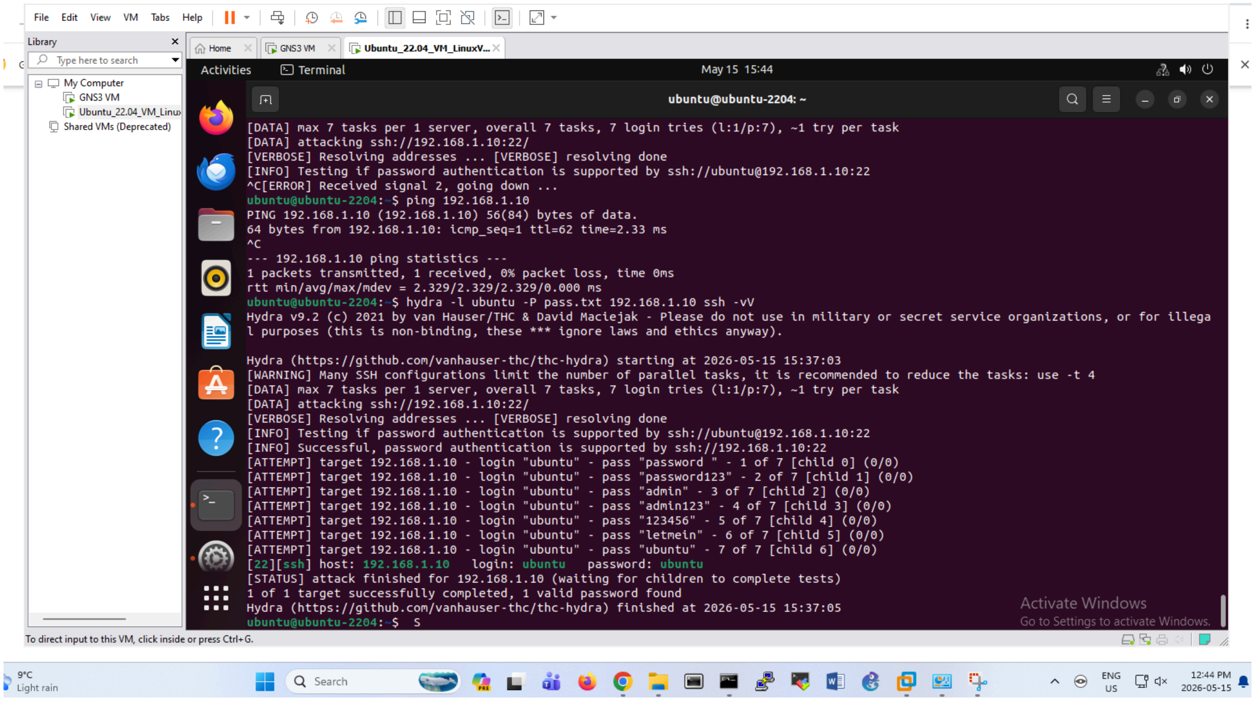
DOS policy


Edit Policy
Name
DOS policy
Incoming Interface
FG1 to FG2
Outgoing Interface
internal
Source
Attacker IP
Destination
internal
Schedule
always
Service
ALL
Action
ACCEPT
DENY
Log Violation Traffic
Comments
Write a comment...
0/1023
Enable this policy
Statistics (since last reset)
ID
7
Last used
1 second(s) ago
First used
11 minute(s) ago
Active sessions
0
Hit count
70
Total bytes
5.64 kB
Current bandwidth
0 bps
Clear Counters
Last 7 Days
Bytes
8 kB
6 kB
4 kB
2 kB
0 B
May 08
May 09
May 10
May 11
Additional Information
API Preview
Edit in CLI
Documentation
Online Help


Google Chrome isn't your default browser
Set as default
splunk>enterprise
Apps
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Report
Logs for FG
Edit
Export
i
Time
Event
5/15/26
12:16:29.000 PM
5/15/26
12:16:27.000 PM
5/15/26
12:12:06.000 PM
host = 192.168.1.99
source = udp:514
sourcetype = fortigate_traffic
Activate Windows
Go to Settings to activate Windows.

ATTACK Simulation (SSH and NMAP)

logs_for_fg1_skils-2026-05-15.pdf



APACHE server